

PUB LOG Dataset Priority for the Capstone Project

The capstone project does not need to load every PUB LOG CSV file. Use the following dataset priority.

Priority 1 Item Identification

Load:

- `P_FLIS_NSN.CSV`

Use it for:

- NIIN
- FSC
- INC
- Item name
- End-item name
- Cancelled-item reference

This is the main item table for the project.

Priority 2 Item Indicators

Load:

- `V_FLIS_IDENTIFICATION.CSV`

Use it for supporting indicators such as:

- Precious Metals Indicator Code
- Criticality code
- Hazardous-material indicators
- Hazardous-characteristics code
- Environmental attributes
- Demilitarization code

These codes provide clues and warnings. They do not prove that a particular material is present.

Priority 3 Material Characteristics

Load:

- V_CHARACTERISTICS.CSV

Use these fields:

- NIIN
- MRC
- REQUIREMENTS_STATEMENT
- CLEAR_TEXT_REPLY

This is the most important PUB LOG file for material discovery. Search the requirement statement and clear-text reply for explicit references to materials such as:

- Aluminum
- Cobalt
- Copper
- Gold
- Graphite
- Lithium
- Magnesium
- Nickel
- Rare earth materials
- Silver
- Tantalum
- Titanium
- Tungsten
- Zinc
- Alloy
- Coating
- Plating

Filter this file using the selected project NIINs. Do not load all 40-million-plus rows into application memory.

How to Reduce the Three Large Files

The goal is not to randomly select a few NSNs. The goal is to create a smaller, traceable dataset containing items that are both relevant and supported by material evidence.

Use the following three-stage process.

Stage 1: Create a Broad Candidate NIIN List

Start with `P_FLIS_NSN.CSV` because it identifies the items.

1. Read only the fields needed for selection:
 - `FSC`
 - `NIIN`
 - `INC`
 - `ITEM_NAME`
 - `END_ITEM_NAME`
 - Cancellation or replacement fields, when present
2. Preserve `FSC` and `NIIN` as text so leading zeroes are not lost.
3. Exclude cancelled items .
4. FSC categories, such as categories related to batteries, and important items
5. Save the distinct matching `NIIN` values as `candidate_niins` .

The FSC and item-name filter establishes the project area. It does not prove that a critical material is present.

Recommended starting size:

- Start with approximately 500 to 5,000 candidate NIINs.
- If fewer than 100 are found, broaden the FSC or item-name selection.
- If more than 10,000 are found, narrow the categories before processing the characteristics.

Stage 2: Reduce the 40-Million-Row Characteristics File

Process `V_CHARACTERISTICS.CSV` as a stream or in chunks.

For each chunk:

1. Read only:
 - `NIIN`
 - `MRC`
 - `REQUIREMENTS_STATEMENT`
 - `CLEAR_TEXT_REPLY`
2. Keep rows whose `NIIN` exists in `candidate_niins` .

3. Search `REQUIREMENTS_STATEMENT` and `CLEAR_TEXT_REPLY` using the USGS critical-material names and approved aliases.
4. Write matching rows immediately to a reduced output file or database table.
5. Discard the processed chunk before reading the next one.

The application must never hold all 40-million-plus rows in memory. It only holds one chunk and the relatively small candidate-NIIN set.

Save every match with its evidence:

- `NIIN`
- `MRC`
- `CRITICAL_MATERIAL`
- `MATCHED_TERM`
- `REQUIREMENTS_STATEMENT`
- `CLEAR_TEXT_REPLY`

After all chunks have been processed, create a distinct list of the NIINs that produced at least one material match. Save this as `selected_niins`.

Stage 3: Reduce All Three Files Using the Selected NIINs

Use `selected_niins` to create the final capstone extracts:

1. From `P_FLIS_NSN.CSV`, keep only rows whose NIIN is in `selected_niins`.
2. From `V_FLIS_IDENTIFICATION.CSV`, keep only rows whose NIIN is in `selected_niins`.
3. From `V_CHARACTERISTICS.CSV`, keep the matching evidence rows written during Stage 2. Other characteristics for the selected NIINs may be retained.

Recommended outputs:

- `selected_niins.csv` — one row per selected NIIN
- `selected_items.csv` — item name, FSC, INC, and NIIN
- `selected_identification.csv` — supporting codes for selected NIINs
- `selected_material_evidence.csv` — material matches and source text

Create the 13-digit NSN only after the join:

```
NSN = four-character FSC + nine-character NIIN
```

Selection Logic Summary

P_FLIS_NSN

- > choose project FSC categories and/or item-name keywords
- > candidate_niins (about 500 to 5,000)

V_CHARACTERISTICS (40M+ rows)

- > read one chunk at a time
- > keep candidate NIINS
- > keep rows containing a USGS material name or alias
- > selected_niins plus material evidence

selected_niins

- > reduce P_FLIS_NSN
- > reduce V_FLIS_IDENTIFICATION
- > retain matching V_CHARACTERISTICS evidence
- > produce the small capstone dataset

Required Validation

- Count the input rows, candidate NIINS, material matches, distinct selected NIINS, and final output rows.
- Manually review a sample of the matches to identify false positives.
- Keep the matched source text so every material decision can be explained.
- Treat material matches as candidate evidence, not as an authoritative bill of materials.
- Do not treat CRIT_CD , PMIC , DMIL , HMIC , or HCC as proof that a specific USGS critical material is present.
- Absence of a material term does not prove that an item contains no critical material; PUB LOG does not provide a complete bill of materials.

Priority 4 Classification Lookups

Load:

- V_H2_FSG.CSV
- V_H2_FSC.CSV
- V_H6_NAME_INC.CSV

Use them to:

- Translate FSC, FSG, and INC codes into readable names.
- Group related items.
- Select relevant product categories.
- Validate classification codes.

Classification lookups describe the type of item. They do not establish its material composition.

Priority 5 Critical Materials Reference

Continue using the current USGS Critical Minerals List.

Use it to:

- Normalize critical-material names.
- Store common aliases and spelling variations.
- Compare materials identified in PUB LOG with the official list.
- Record the list year and source URL for reproducibility.

Recommended Workflow

P_FLIS_NSN

- > select project categories
- > create candidate_niins
- > scan V_CHARACTERISTICS in chunks
- > filter by candidate NIIN and USGS material terms
- > create selected_niins and retain evidence
- > filter P_FLIS_NSN and V_FLIS_IDENTIFICATION by selected_niins
- > send uncertain results for human review